SOFTWARE REQUIREMENTS SPECIFICATION

AI-Powered Contract Intelligence & Clause Extraction Platform

Detailed System, Functional and Data Requirements for the C.L.E.A.R. Contract Intelligence Platform, Derived from BRD v1.1

Domain	Category	Priority	Status
Legal & Secretarial	ROI / Productivity Enhancement	High	Draft

Version	Prepared For	Prepared By	Date
2.0	Hackathon Evaluation	Business Analyst	01 Aug 2026

DOCUMENT SUMMARY

This Software Requirements Specification translates the business need and scope defined in the C.L.E.A.R. Business Requirements Document (v1.1) into detailed, verifiable functional, data, interface and non-functional requirements for design and development. Version 2.0 adds full coverage of system architecture and technology stack, deployment mechanism and CI/CD, the ER diagram and data model, detailed non-functional requirements (performance, scalability, availability, usability, security, maintainability), security considerations (authentication, authorization, data protection, secrets management, threat mitigation), and the key design considerations and trade-offs behind the architecture.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, data, interface, and non-functional requirements for C.L.E.A.R., an AI-powered Contract Intelligence Platform. It expands the business need and scope captured in the Business Requirements Document (BRD v1.1) into requirements that are specific, measurable and testable, so that they can be used directly for solution design, development and quality assurance.

> **WHAT THIS MEANS**
>
> Where the BRD describes what the business needs and why, this SRS describes what the system must do, how it must behave, and what data it must store, so that developers and testers can build and verify the platform without ambiguity.

1.2 Scope

The system in scope for this SRS covers project-wise contract ingestion, AI-based document parsing and classification, clause extraction and mapping, risk and missing-clause detection, semantic and keyword search, contract snapshot and AI insights dashboards, administration and Clause Master configuration, and export/reporting capabilities, as defined in Sections 3 and 4 of the BRD.

Contract drafting, negotiation/redlining workflows, e-signature capability, external CLM integration, legal interpretation, and multi-language clause extraction remain out of scope for this phase, consistent with BRD Section 3.2.

1.3 Intended Audience

- Solution architects and developers responsible for building the AI processing pipeline, database and application layers.
- QA and test engineers who will design test cases against the acceptance criteria in Section 9.
- Legal, Procurement, Finance, Compliance and Business stakeholders who will validate that requirements meet operational needs.
- Project sponsors and reviewers evaluating solution readiness for Phase 1 delivery.

1.4 Document Conventions

- Functional requirements are identified as FR-<module>-<number> (e.g., FR-DOC-01).
- Non-functional requirements are identified as NFR-<number>.
- Acceptance criteria are identified as AC-<number>, consistent with BRD Section 12 numbering.
- The keywords "must", "should" and "may" indicate mandatory, recommended and optional requirements respectively.

1.5 References

- Business Requirements Document — C.L.E.A.R. AI-Powered Contract Intelligence & Clause Extraction Platform, v1.1.
- Clause Master configuration (referenced in BRD Section 8.3) governing mandatory clauses and risk scoring.

2. Overall Description

2.1 Product Perspective

C.L.E.A.R. is a new, standalone contract intelligence platform. It is not a replacement for existing document storage but sits alongside project repositories, ingesting agreements and producing a structured, searchable intelligence layer on top of them. The system is composed of a document ingestion and management layer, an AI processing pipeline (OCR/parsing, classification, clause extraction, embeddings), a relational data store (Document Master, Document Content Master, Clause Master), and a presentation layer (dashboards, search, export).

2.2 Product Functions (Summary)

Function Area	Summary
Document Management	Project-wise repository with single/bulk ingestion, metadata, versioning and status tracking.
AI Processing	OCR/parsing, JSON generation, document-type classification, clause extraction, chunking, embeddings and summarization.
Clause Intelligence	Clause mapping to Clause Master, page-level traceability, risk scoring and missing-clause detection.
Search & Analytics	Keyword, clause, semantic and repository search with dashboard insights.
Dashboards	Contract Snapshot and AI Insights views for fast business decision-making.
Administration	User roles, project management and Clause Master configuration.
Reporting	Export of summaries, clause data, risk reports and contract inventory.

2.3 User Classes and Characteristics

User Class	Characteristics / Primary Use
Legal Head	Business sponsor and governance owner; approves the clause interpretation framework; occasional, high-privilege use.
Legal Team	Reviews AI output, validates clauses, corrects mappings and approves summaries; frequent, detail-oriented use.

User Class	Characteristics / Primary Use
Procurement Team	Searches vendor agreements for tenure, payment terms, termination and renewal detail; search-focused use.
Finance Team	Reviews contract value, payment obligations, liability and closure terms; read and export-focused use.
Business Teams	Submit contracts and consume snapshots/insights; light, occasional use.
Compliance Team	Monitors risk indicators, missing clauses and governance exceptions; dashboard-focused use.
System Administrator	Manages projects, users, roles and Clause Master configuration; technical/administrative use.

2.4 Operating Environment

- Web-based application accessible via standard modern browsers over the organizational network.
- Backend AI processing services (OCR/ADI, LLM classification, embedding generation) operating as internal or cloud-hosted services.
- Relational database hosting Document Master, Document Content Master and Clause Master, with a vector-capable store or extension for embeddings.

2.5 Design and Implementation Constraints

- Must support PDF, scanned PDF, DOC and DOCX input formats only for Phase 1 (BRD Section 3.1).
- Must not include contract drafting, negotiation, e-signature or external CLM integration in Phase 1 (BRD Section 3.2).
- Clause detection logic must fall back to four-page chunking when section headings are not available (BRD Section 7.1).
- All role-based access, storage and audit logging must comply with organizational data privacy and retention policy.

2.6 Assumptions and Dependencies

- Users are provisioned with valid organizational credentials and role assignments prior to platform access.
- A Clause Master configuration with an initial set of mandatory clauses, keywords and risk weights is available before go-live.
- Underlying OCR/ADI and LLM services are available with acceptable latency and accuracy for legal documents.

3. System Features (Functional Requirements)

3.1 Document Ingestion & Management

FR-DOC-01 — Project-wise Document Submission

The system must allow users to submit one or more contract documents under a selected project or repository category.

Attribute	Detail
Inputs	Single file or bulk file batch; selected Project; supported formats (PDF, scanned PDF, DOC, DOCX).
Processing	Validate file format, file size, duplicate status and user access rights before acceptance.
Outputs	Document record created in Document Master with ProcessingStatus = Pending; confirmation to user.
Priority	High
Related Acceptance Criteria	AC-01, AC-02

FR-DOC-02 — Metadata, Versioning and Status Tracking

The system must maintain document-level metadata, version history and both processing and review status for every submitted contract.

Attribute	Detail
Inputs	Document metadata (name, path, uploader, timestamps).
Processing	Persist and update ProcessingStatus (Pending, In Progress, Processed, Failed) and ReviewStatus (Pending Review, Reviewed, Exception).
Outputs	Up-to-date Document Master record queryable by project, type, status and date.
Priority	High

3.2 AI-Based Document Parsing

FR-AI-01 — OCR / Layout Parsing

The system must extract text, layout, tables, headings and page references from each submitted document, including scanned PDFs.

Attribute	Detail
Inputs	Source document (PDF, scanned PDF, DOC, DOCX).
Processing	ADI/OCR engine parses structure and produces a structured JSON representation (title, section headings, tables, paragraphs, page numbers).

Attribute	Detail
Outputs	JSON output persisted and referenced via JsonPath in Document Master.
Priority	High
Related Acceptance Criteria	AC-03

3.3 Document Type Classification

FR-AI-02 — LLM-Based Document Type Identification

The system must automatically identify the contract type (e.g., NDA, MSA, SLA, Contract Cum Order Form, Licensing Agreement) using an LLM.

Attribute	Detail
Inputs	Parsed JSON structure — title, section headings, key phrases and legal context.
Processing	LLM classifies the document against the set of predefined document types.
Outputs	DocumentType field populated in Document Master; classification available for override by a Legal Reviewer or Admin.
Priority	High
Related Acceptance Criteria	AC-04

3.4 Clause Extraction & Mapping

FR-CL-01 — Header-Based Clause Detection

Where meaningful section headers exist, the system must identify clauses using the document type and section context.

Attribute	Detail
Inputs	Parsed JSON with section headings; identified DocumentType.
Processing	LLM identifies clauses per section and maps them against Clause Master definitions.
Outputs	Clause records created in Document Content Master with ClauseName, ClauseType and SectionHeader.
Priority	High

Attribute	Detail
Related Acceptance Criteria	AC-05

FR-CL-02 — Chunk-Based Clause Discovery

Where section headers are not available, the system must split content into four-page chunks and perform semantic clause discovery on each chunk.

Attribute	Detail
Inputs	Parsed JSON without usable section headings.
Processing	Content split into four-page chunks; LLM performs semantic discovery of clauses within each chunk.
Outputs	Clause records tagged with ChunkId in Document Content Master.
Priority	High
Related Acceptance Criteria	AC-06

FR-CL-03 — Clause Mapping, Traceability and Confidence

Every extracted clause must be mapped to the Clause Master, include a page-level reference to its source, and carry an AI confidence score.

Attribute	Detail
Inputs	Extracted clause text and its originating page/section.
Processing	Map clause to ClauseId in Clause Master; capture ReferencePage, ConfidenceScore and source Content.
Outputs	Clause record in Document Content Master displaying source content and page number to the user.
Priority	High
Related Acceptance Criteria	AC-07, AC-08, AC-09

3.5 Risk & Missing Clause Detection

FR-RISK-01 — Risk Scoring and Missing Clause Flags

The system must identify missing mandatory clauses, high-risk terms and non-standard language for each processed contract.

Attribute	Detail
Inputs	Extracted clauses; Clause Master MandatoryFlag and RiskWeight configuration; DocumentType.
Processing	Compare extracted clauses against mandatory clause list per document type; compute RiskLevel and MissingClauseFlag.
Outputs	RiskLevel (Low/Medium/High) and MissingClauseFlag stored per clause; high-risk and missing clauses surfaced on the Contract Snapshot and dashboards.
Priority	High
Related Acceptance Criteria	AC-11

3.6 Search & Analytics

FR-SEARCH-01 — Keyword, Clause and Semantic Search

Users must be able to search contracts and clauses by clause, counterparty, date, document type, value or keyword, including semantic (embedding-based) matches.

Attribute	Detail
Inputs	Search query (structured filters and/or free-text keyword).
Processing	Query executed against Document Master and Document Content Master; semantic queries matched against clause embeddings.
Outputs	Ranked list of matching documents/clauses with navigation to source page.
Priority	High
Related Acceptance Criteria	AC-12

3.7 Dashboards

FR-DASH-01 — Contract Snapshot

The system must present a business-friendly snapshot per contract covering type, value, key terms, important dates, clause overview and an AI-generated summary.

Attribute	Detail
Inputs	Document Master and Document Content Master records for the selected contract.
Processing	Aggregate contract information, key terms, dates and clause counts; retrieve AI-generated summary.
Outputs	Contract Snapshot view (see Section 6 for detailed layout).
Priority	High
Related Acceptance Criteria	AC-10

FR-DASH-02 — AI Insights Dashboard

The system must surface Risk, Renewal, Business and Review insights across the contract repository.

Attribute	Detail
Inputs	Aggregated clause, risk and review data across projects.
Processing	Compute insight categories (risk exposure, upcoming renewals, business obligations, review needs).
Outputs	AI Insights Dashboard view, filterable by project and document type.
Priority	Medium

3.8 Administration & Configuration

FR-ADM-01 — User, Role and Project Administration

Administrators must be able to manage projects, users and role assignments controlling document, dashboard and configuration access.

Attribute	Detail
Inputs	User, role and project definitions.
Processing	Create/update/deactivate users, roles and project mappings.
Outputs	Role-based access enforced across the platform.
Priority	High

FR-ADM-02 — Clause Master Configuration

Administrators must be able to configure mandatory clauses, keywords, definitions, risk weights and document-type applicability, with version control.

Attribute	Detail
Inputs	ClauseName, ApplicableDocType, Keywords, Description, MandatoryFlag, RiskWeight, ActiveFlag.
Processing	Persist configuration with an incremented VersionNo for traceability.
Outputs	Updated Clause Master used by clause extraction and risk-scoring logic.
Priority	High

3.9 Reporting & Export

FR-EXP-01 — Export of Summaries, Clauses and Reports

The system must allow export of contract summaries, extracted clause data, risk reports and search results.

Attribute	Detail
Inputs	Selected contract(s), clause set, or search/report result set.
Processing	Format selected data for export (e.g., document or spreadsheet format).
Outputs	Downloadable export file.
Priority	Medium
Related Acceptance Criteria	AC-13

3.10 Legal Validation Workflow

FR-VAL-01 — Review, Correction and Override of AI Output

Legal reviewers must be able to accept, correct, comment on or override AI-generated clause extraction, classification and summaries.

Attribute	Detail
Inputs	AI-extracted clause data, document type, and summary for a given contract.
Processing	Capture reviewer action (accept/correct/override/comment) and update ReviewStatus.
Outputs	Finalized, reviewer-validated clause and document records.

Attribute	Detail
Priority	High
Related Acceptance Criteria	AC-14

4. External Interface Requirements

4.1 User Interfaces

- A web-based interface providing project selection, document submission, Contract Snapshot, Clause Analysis, Search Repository and AI Insights Dashboard views.
- A simple, business-friendly design usable by both legal and non-legal users, consistent with BRD Section 11 (Usability).
- An administration console for project, user, role and Clause Master management.

4.2 Hardware Interfaces

- No specialized hardware is required; standard client devices (desktop/laptop) with a modern browser are sufficient.

4.3 Software Interfaces

Interface	Purpose
OCR / Document Intelligence engine	Extracts text, layout, tables and page references from source documents.
LLM service	Performs document classification, clause identification, summarization and insight generation.
Relational database	Persists Document Master, Document Content Master and Clause Master.
Vector store / embedding index	Stores clause embeddings and serves semantic search queries.
Export/reporting engine	Generates downloadable summaries, clause exports and risk reports.

4.4 Communication Interfaces

- Standard HTTPS-based communication between the client interface and backend services.
- Secure, authenticated service-to-service calls between the application layer and AI processing/search services.

5. System Architecture

5.1 High-Level Architecture

The platform is organized into five layers plus a cross-cutting concerns band, as shown in Figure 1. Contracts enter through the client layer, pass through the API and authentication layer (backed by the organization's identity provider via SSO), are orchestrated by the application services, processed by the AI layer, and persisted in the data layer. Secrets management and observability run across every layer rather than sitting in any single one.

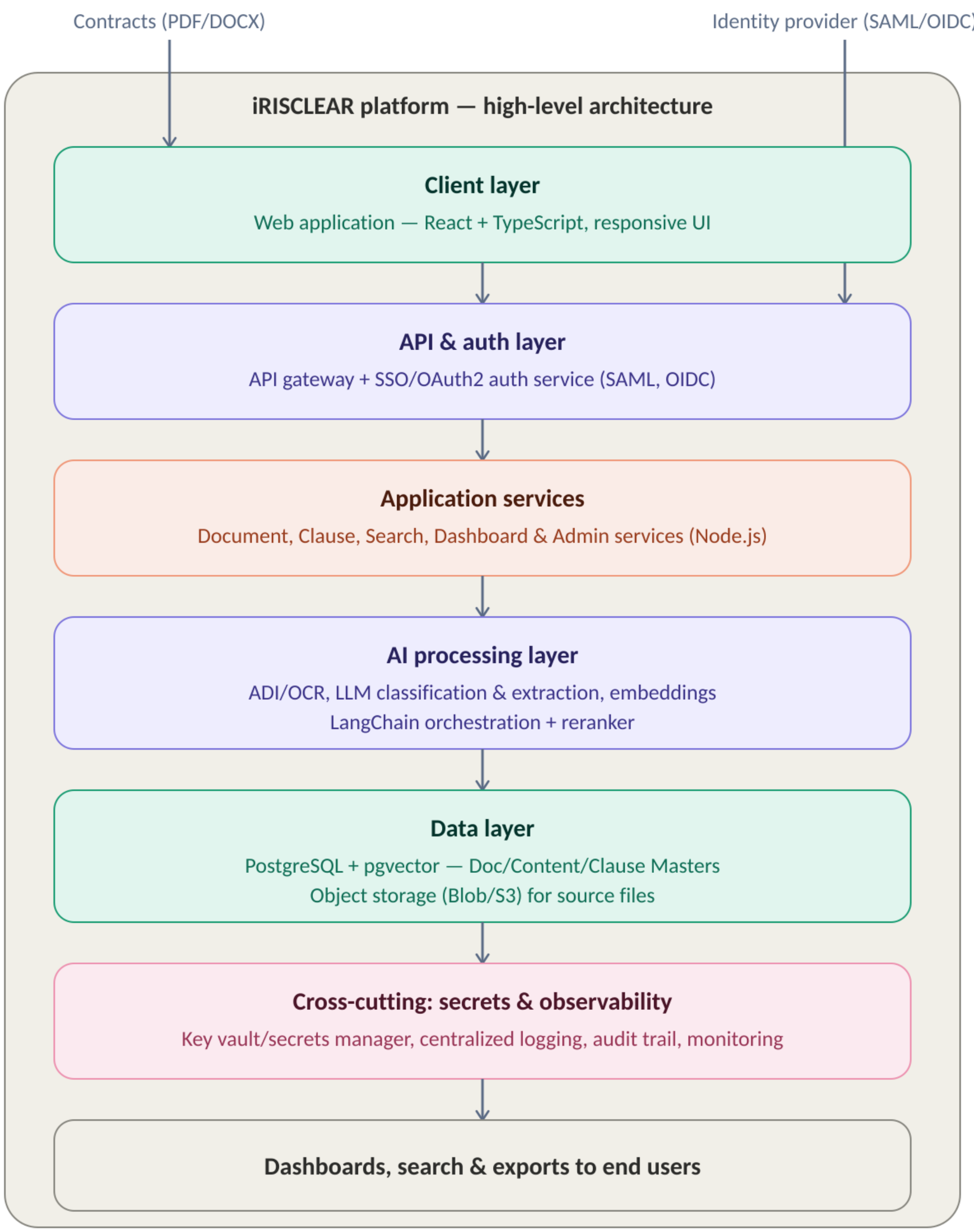

Figure 1 — iRISCLEAR high-level architecture, client layer through data layer.

5.2 Component Descriptions

Component	Responsibility
Web application (client layer)	Renders project selection, document submission, Contract Snapshot, Clause Analysis, Search and AI Insights views.
API gateway	Single entry point for all client requests; routes to application services, enforces rate limiting and request validation.
Auth service (SSO/OAuth2)	Delegates authentication to the organization's identity provider (SAML/OIDC) and issues session tokens; enforces role-based authorization.
Document service	Handles submission, metadata, versioning and processing-status tracking for contracts (Doc Master).
Clause service	Orchestrates clause extraction, mapping to Clause Master, risk scoring and missing-clause detection.
Search service	Serves keyword, clause and semantic search queries against Document Content Master and embeddings.
Dashboard service	Aggregates data for Contract Snapshot and AI Insights views.
Admin service	Manages projects, users, roles and Clause Master configuration.
AI processing layer	Runs ADI/OCR parsing, LLM document-type classification, LLM clause extraction, embedding generation and the LangChain-based query pipeline (including reranking).
Data layer	PostgreSQL with the pgvector extension for relational and vector data; object storage for source PDFs/DOCX and generated JSON.
Secrets & observability	Centralized secrets manager/key vault, structured logging, audit trail and monitoring/alerting across all services.

5.3 Technology Stack

Layer	Technology Choices
Frontend	React with TypeScript; component library for dashboards, tables and forms; responsive layout for desktop and tablet use.
API / backend services	Node.js (or equivalent managed runtime) exposing REST/JSON APIs behind the API gateway; service-to-service calls secured with short-lived tokens.
Authentication	SSO via SAML 2.0 or OpenID Connect against the organization's identity provider; OAuth2 for service tokens.

Layer	Technology Choices
AI / ML	Document Intelligence (ADI/OCR) engine; LLM for classification, clause extraction and answer generation; embedding model for vector representations; LangChain for pipeline orchestration; a reranking model for retrieval quality.
Database	PostgreSQL with the pgvector extension, hosting Document Master, Document Content Master and Clause Master, including vector columns for semantic search.
Storage	Object storage (Blob/S3-compatible) for original contract files and parsed JSON output.
Secrets management	Managed key vault / secrets manager for credentials, API keys and connection strings.
CI/CD & hosting	Containerized services (Docker) deployed via a CI/CD pipeline to a cloud environment (see Section 6).
Observability	Centralized logging, application performance monitoring and audit-log storage.

5.4 Document Processing Flow

Within the AI processing layer, a submitted document moves through the following stages to become structured, searchable intelligence.

Stage	Description
Document	Source legal agreement available for backend processing.
ADI Processing	OCR and parsing engine extracts text, layout, tables, headings and page references.
JSON Output	Structured representation of title, section headings, tables, paragraphs and page numbers.
LLM Document Type Identification	LLM identifies the contract type based on title, sections, key phrases and legal context.
Section Heading Check	System checks whether meaningful section headers or titles are available.
Header-Based Clause Detection	If headings exist, LLM identifies clauses using document type and section context.
4-Page Chunking	If headings do not exist, content is split into four-page chunks for semantic clause discovery.
Clause Mapping	Identified clauses are mapped to the Clause Master and tagged with confidence and risk indicators.
Embedding Generation	Clause text and chunks are converted into vectors to support semantic search and AI retrieval.

Stage	Description
Data Persistence	Document-level metadata is stored in Document Master; clause content is stored in Document Content Master.
Dashboard & Search	Contract Snapshot, AI Insights, searchable repository and reports are generated.

PROCESSING PRINCIPLE

The backend flow begins with Document, not Document Upload. Upload is a user action, while Document is the starting point of internal AI processing.

6. Deployment Details & Mechanism

6.1 Environments

Environment	Purpose
Development (Dev)	Used by engineers for active feature development and unit testing; connected to sandboxed AI services and a non-production database.
Staging / QA	Mirrors production configuration for integration testing, UAT and performance testing before release.
Production (Prod)	Live environment serving end users, with production-grade monitoring, backups and access controls.

6.2 Deployment Steps

- 1. Code is committed to the version control repository on a feature branch and merged via pull request after review.
- 2. The CI pipeline builds the application, runs automated unit and integration tests, and produces versioned container images.
- 3. Container images are pushed to a container registry and scanned for known vulnerabilities.
- 4. The CD pipeline deploys the new images to the Dev environment automatically, and to Staging after CI checks pass.
- 5. Database schema changes are applied through versioned migration scripts as part of the pipeline, never manually against production.
- 6. Promotion to Production requires a manual approval gate and runs as a rolling or blue-green deployment to avoid downtime.
- 7. Post-deployment smoke tests verify core flows (login, document submission, search) before the release is marked complete.

6.3 CI/CD Approach

Stage	Description
Source control	Trunk-based development with short-lived feature branches; pull requests required for merges to main.
Continuous Integration	Automated build, lint, unit tests and integration tests triggered on every pull request and merge.
Artifact packaging	Application and AI processing services are packaged as versioned Docker images.
Continuous Deployment	Automated deployment to Dev and Staging; gated, approval-based deployment to Production.
Infrastructure as Code	Environment configuration (network, database, secrets references) managed as code for repeatable provisioning.
Rollback	Previous container image version and database migration rollback scripts retained for fast recovery.

DEPLOYMENT PRINCIPLE

No environment is configured by hand. Every environment difference (secrets, endpoints, feature flags) is expressed as configuration, not as a code change, so the same build artifact can be promoted from Dev through to Production unchanged.

7. Data Model / ER Diagram

7.1 Entity Relationship Diagram

The data model centers on three contract-level tables (Doc Master, Doc Content Master, Clause Master) and two access-control tables (User, Role), all scoped by Project. Figure 2 shows the relationships between them.

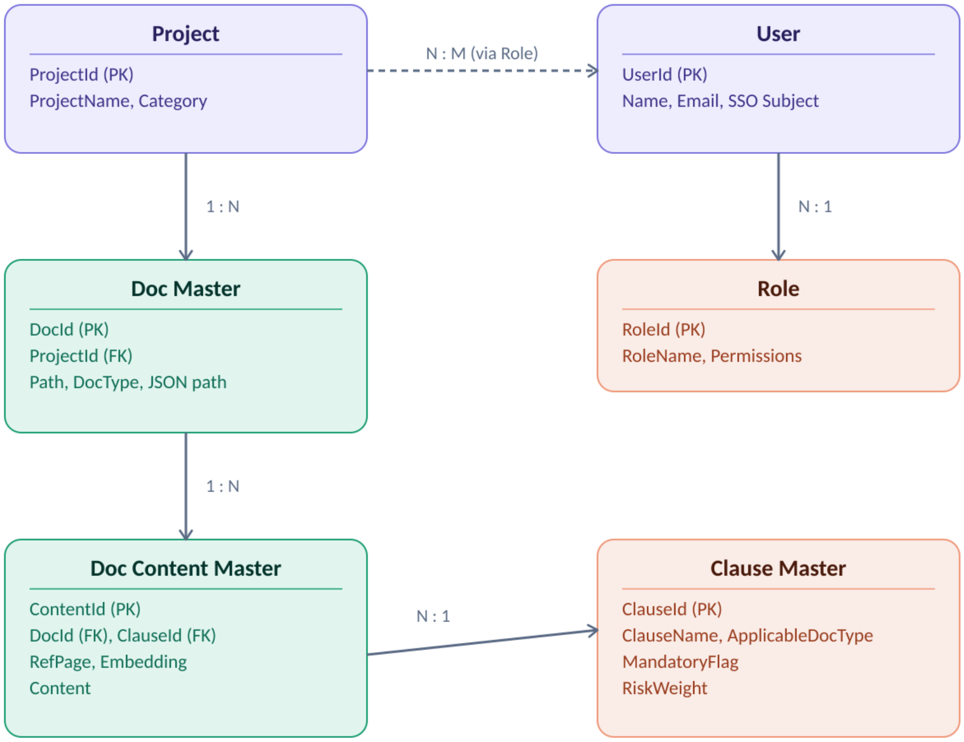

Figure 2 — Core entity relationships: Project, User, Role, Doc Master, Doc Content Master and Clause Master.

7.2 Entity Relationships

Relationship	Cardinality	Description
Project → Doc Master	1 : N	A project can contain many documents; every document belongs to exactly one project.
Doc Master → Doc Content Master	1 : N	A document can yield many extracted clause/content records.
Doc Content Master → Clause Master	N : 1	Many extracted content records map to the same clause definition in Clause Master.
Role → User	1 : N	A role can be assigned to many users; each user has one active role in a given context.
User ↔ Project	N : M	Users can have access to multiple projects, and a project can have multiple users, mediated by role assignment.

7.3 Doc Master

Doc Master is the document-level database table. It stores the core identity, location, classification, processing and review status for each contract.

Column	Data Type	Description
DocId	Unique Identifier (PK)	Unique system-generated document identifier.
ProjectId	Unique Identifier (FK)	Project or repository category linked to the document.
DocumentName	Text	Original uploaded file name.
DocumentPath	Text (path/URI)	Storage path of the source document.
JsonPath	Text (path/URI)	Storage path of the parsed JSON output.
DocumentType	Text (enumerated)	Classified contract type such as NDA, MSA, SLA or Vendor Agreement.
CounterpartyName	Text	Vendor, client or other contracting party extracted from the document.
ContractValue	Decimal / Currency	Monetary value extracted from the document, where available.
EffectiveDate	Date	Contract start date.
ExpiryDate	Date	Contract end date.
ContractTerm	Text / Duration	Duration or tenure of the agreement.
ProcessingStatus	Text (enumerated)	Current AI processing status: Pending, In Progress, Processed or Failed.
ReviewStatus	Text (enumerated)	Business review status: Pending Review, Reviewed or Exception.
UploadedBy / UploadedDate	Text / Timestamp	User and timestamp for document submission.
CreatedOn / ModifiedOn	Timestamp	System timestamps for record creation and updates.

7.4 Doc Content Master

Doc Content Master stores the clause-level knowledge extracted from each document. It supports semantic search, clause dashboards, evidence traceability and AI retrieval.

Column	Data Type	Description
ContentId	Unique Identifier (PK)	Unique clause or content record identifier.

Column	Data Type	Description
DocId	Unique Identifier (FK)	Foreign key linking content to Doc Master.
ClauseId	Unique Identifier (FK)	Foreign key linking to the clause definition in Clause Master.
ClauseName	Text	Extracted clause name, such as Termination, Liability or Payment Terms.
ClauseType	Text (enumerated)	Clause category used for reporting and risk analysis.
SectionHeader	Text	Section or title from which the clause was extracted, where available.
ReferencePage	Integer	Page number where the clause appears in the contract.
Content	Long Text	Extracted clause text.
ChunkId	Text / Identifier	Identifier of the four-page chunk if chunk-based processing was used.
Embedding	Vector (float array)	Vector representation used for semantic search.
ClauseSummary	Text	AI-generated explanation of the clause.
RiskLevel	Text (enumerated)	Risk indicator: Low, Medium or High.
MissingClauseFlag	Boolean	Indicates whether a mandatory clause was not found.
ConfidenceScore	Decimal (0–1 or %)	AI confidence score for extraction accuracy.

7.5 Clause Master

Clause Master defines what the system should identify, validate and monitor. Admins configure mandatory clauses, keywords, definitions, risk scores and document-type applicability here.

Field	Data Type	Description
ClauseId	Unique Identifier (PK)	Unique clause identifier.
ClauseName	Text	Clause or key term title.
ApplicableDocType	Text (enumerated / list)	Document type where the clause is expected or relevant.
Keywords	Text / List	Keywords, patterns or phrases used for detection.
Description	Text	Business/legal definition of the clause.
MandatoryFlag	Boolean	Defines whether the clause is mandatory for a document type.

Field	Data Type	Description
RiskWeight	Integer / Decimal	Risk score used to calculate clause or contract risk.
ActiveFlag	Boolean	Indicates whether the clause is currently active for extraction.
VersionNo	Integer	Configuration version used for traceability.

7.6 User & Role (Access Control)

Table	Key Columns	Description
User	UserId (PK), Name, Email, SSOSubject, ActiveFlag	Identity record synchronized from the organization's identity provider on first SSO login.
Role	RoleId (PK), RoleName, Permissions	Defines a named permission set (e.g., Legal Reviewer, Admin, Viewer) assignable to users per project.
UserProjectRole	UserId (FK), ProjectId (FK), RoleId (FK)	Join table granting a user a specific role within a specific project, enabling the N:M user-to-project relationship.

7.7 Sample Data Structure

The following illustrates the shape of a single Doc Content Master record as it would be produced by the AI processing pipeline and consumed by search and dashboards.


```
{
  "ContentId": "c-8f21",
  "DocId": "d-4471",
  "ClauseId": "cl-012",
  "ClauseName": "Termination for Convenience",
  "ClauseType": "Termination",
  "SectionHeader": "9. Termination",
  "ReferencePage": 14,
  "Content": "Either party may terminate this Agreement for convenience upon 30 days written notice...",
  "ChunkId": null,
  "Embedding": [0.0123, -0.0871, 0.0456, "... 1536 dims"],
  "ClauseSummary": "Allows either party to end the contract with 30 days notice, without cause.",
  "RiskLevel": "Medium",
  "MissingClauseFlag": false,
  "ConfidenceScore": 0.94
}
```


8. Dashboard Requirements

8.1 Contract Snapshot

PURPOSE

The Contract Snapshot gives users a quick business-friendly view of a contract without reading the full document.

Snapshot Area	Data Displayed
Contract Information	Document Type, Counterparty, Contract Value, Status and Project.
Key Terms	Payment Terms, Governing Law, Auto Renewal, Notice Period and Liability Cap.
Important Dates	Effective Date, Expiry Date, Renewal Date, Notice Date and Contract Duration.
Clause Overview	Total clauses found, missing clauses, mandatory clauses and high-risk clauses.
Summary	Concise AI-generated explanation of the agreement and its major obligations.

8.2 AI Insights Dashboard

Insight Category	Description
Risk Insights	Highlights high-risk clauses, unusual language, missing protections and liability exposure.
Renewal Insights	Highlights upcoming renewals, auto-renewal terms and expiring agreements.
Business Insights	Summarizes payment obligations, commercial commitments, vendor dependencies and key legal obligations.
Review Insights	Identifies contracts requiring legal review, low-confidence extractions and clauses needing validation.

8.3 Sample Dashboard API Response

The following is an actual response returned by the dashboard/insights endpoint at application scope, showing the KPI values, distributions, top risks and upload trend that populate the Contract Snapshot and AI Insights views.


```
{
  "scope": "application",
  "project_ids": [
    "0f0909da-fb6b-4983-9ab3-ceed67a5794c",
    "de5cfc5b-af03-42a3-8700-cf7ddcf2f0db",
    "2c82c027-3038-428b-bfc5-ccfdd4488a5b"
  ],
  "kpis": [
    {
      "key": "total_contracts",
      "label": "Contracts",
      "value": 5,
      "delta": null,
```

```
      "unit": null,
      "drilldown": null
    },
    {
      "key": "expiring_soon",
      "label": "Expiring in 90 days",
      "value": 0,
      "delta": null,
      "unit": null,
      "drilldown": {
        "expiring_before": "2026-10-30",
        "expiring_after": "2026-08-01"
      }
    },
    {
      "key": "high_risk",
      "label": "High risk",
      "value": 0,
      "delta": null,
      "unit": null,
      "drilldown": {
        "risk_band": "high"
      }
    },
    {
      "key": "unlimited_liability",
      "label": "Unlimited liability",
      "value": 0,
      "delta": null,
      "unit": null,
      "drilldown": {
        "has_unlimited_liability": true
      }
    },
    {
      "key": "missing_clauses",
      "label": "Missing mandatory clauses",
      "value": 2,
      "delta": null,
      "unit": null,
      "drilldown": {
        "missing_mandatory": true
      }
    },
    {
      "key": "needs_review",
      "label": "Needs review",
      "value": 3,
      "delta": null,
      "unit": null,
      "drilldown": {
        "needs_review": true
      }
    },
    {
      "key": "total_value",
      "label": "Total value",
      "value": 0,
      "delta": null,
      "unit": "USD",
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```
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}
```


READING THIS RESPONSE

kpis carries the seven headline tiles (Contracts, Expiring in 90 days, High risk, Unlimited liability, Missing mandatory clauses, Needs review, Total value), each with an optional drilldown filter used when a user clicks the tile.

risk_distribution, agreement_type_distribution and status_distribution drive the dashboard's chart widgets; top_risks lists the highest-scoring contracts with their severity breakdown for the risk table.

uploads_over_time feeds the upload-trend chart, and generated_at records when this snapshot of the dashboard was computed.

9. Non-Functional Requirements

This section defines measurable, testable non-functional requirements across six categories: performance, scalability, availability, usability, security and maintainability.

9.1 Performance

ID	Requirement
NFR-PERF-01	Interactive UI actions (navigation, filtering, opening a Contract Snapshot) must return within 2 seconds under normal load.
NFR-PERF-02	Keyword and clause search must return results within 3 seconds for a repository of up to 50,000 documents.
NFR-PERF-03	Semantic (RAG) queries must return an answer with sources within 8 seconds end-to-end, including embedding, vector search, reranking and LLM generation.
NFR-PERF-04	Document processing (OCR, classification, clause extraction) for a typical 20-page agreement must complete within 3 minutes.

9.2 Scalability

ID	Requirement
NFR-SCALE-01	The system must support horizontal scaling of application and AI processing services independently, based on load.
NFR-SCALE-02	The data layer must support at least 1 million clause/content records and 100,000 documents without redesign, using appropriate indexing on the vector column.

ID	Requirement
NFR-SCALE-03	The document processing pipeline must support queued, asynchronous processing so that bulk uploads do not block interactive use of the platform.
NFR-SCALE-04	The system must support a growing number of concurrent users and projects without degradation, validated through load testing before major releases.

9.3 Availability

ID	Requirement
NFR-AVAIL-01	The platform must target 99.5% monthly uptime for core services (login, document access, search, dashboards).
NFR-AVAIL-02	Planned maintenance windows must be communicated in advance and scheduled outside core business hours where possible.
NFR-AVAIL-03	The system must degrade gracefully if the AI processing layer is unavailable: previously processed documents, search and dashboards must remain accessible.
NFR-AVAIL-04	Automated backups of the database and object storage must run on a defined schedule, with a documented recovery point objective (RPO) and recovery time objective (RTO).

9.4 Usability

ID	Requirement
NFR-USE-01	The interface must remain simple and business-friendly for both legal and non-legal users, requiring minimal training.
NFR-USE-02	Every AI-generated output (clause, summary, risk flag) must be traceable to its source page, so users are never asked to trust an unexplained answer.
NFR-USE-03	The platform must be usable on standard desktop and tablet screen sizes without horizontal scrolling for core workflows.
NFR-USE-04	Error and validation messages must be written in plain business language, not raw system or stack-trace text.

9.5 Security

ID	Requirement
NFR-SEC-01	The system must enforce role-based access control and Single Sign-On (SSO) based authentication for all user access.
NFR-SEC-02	All contract data must be encrypted at rest and in transit.
NFR-SEC-03	The system must maintain a detailed, tamper-evident audit log of uploads, extraction results, reviews and overrides.
NFR-SEC-04	The system must follow organizational data privacy, retention and access-control policies at all times.

Section 10 (Security Considerations) expands on authentication, authorization, data protection, secrets management and threat mitigation in detail.

9.6 Maintainability

ID	Requirement
NFR-MAINT-01	Application and AI processing services must be independently deployable and versioned, avoiding tight coupling between services.
NFR-MAINT-02	Clause Master and other configuration must be editable by administrators without a code deployment, with version history retained.
NFR-MAINT-03	The codebase must maintain automated test coverage for core business logic, enforced as part of the CI pipeline (see Section 6.3).
NFR-MAINT-04	System and integration documentation (API contracts, data model, deployment runbook) must be kept current with each release.

10. Security Considerations

This section expands NFR-SEC-01 through NFR-SEC-04 into concrete mechanisms for authentication, authorization, data protection, secrets management and threat mitigation.

10.1 Authentication

- Users authenticate exclusively through Single Sign-On (SSO), using the organization's identity provider via SAML 2.0 or OpenID Connect — the platform does not maintain its own username/password store.
- On first login, a User record is provisioned from the SSO identity claims (name, email, subject identifier); subsequent logins match on the SSO subject.
- Session tokens are short-lived and refreshed via the identity provider; idle sessions expire automatically after a configurable period of inactivity.
- Service-to-service calls between application services and the AI processing layer use separate, short-lived OAuth2 tokens, distinct from user session tokens.

10.2 Authorization

- Access is role-based (RBAC): each user is assigned one or more roles per project through the UserProjectRole mapping (Section 7.6).
- Roles such as Legal Head, Legal Team, Procurement, Finance, Compliance and Administrator map to specific permissions on documents, dashboards, search, export and Clause Master configuration, consistent with the user classes in Section 2.3.
- Authorization checks are enforced at the API gateway and re-validated in each application service, so that no service trusts the client alone for access decisions.
- Project-level scoping ensures a user can only see and search documents belonging to projects they have been granted access to.

10.3 Data Protection

- Contract data is encrypted in transit using TLS for all client-to-server and service-to-service communication.
- Contract data is encrypted at rest, both in the PostgreSQL database and in object storage holding source files and parsed JSON.
- Personally identifiable or commercially sensitive contract terms are not sent to any external AI service that is not covered by the organization's data processing agreement.
- Data retention and deletion follow the organization's records-management policy; deleted documents are purged from both the database and object storage within the defined retention window.

10.4 Secrets Management

- Database credentials, API keys for AI/OCR services, and identity-provider client secrets are stored in a managed secrets manager / key vault, never in source control or plain configuration files.
- Services retrieve secrets at runtime through a scoped identity, with least-privilege access to only the secrets each service needs.
- Secrets are rotated on a defined schedule and immediately upon suspected compromise, without requiring application code changes.

10.5 Threat Mitigation

Threat	Mitigation
Unauthorized access to contracts	SSO authentication, project-scoped role-based authorization, and audit logging of all access.
Injection attacks (SQL, prompt injection into LLM calls)	Parameterized queries, input validation at the API gateway, and prompt-construction safeguards that separate user input from system instructions in the RAG pipeline.
Data exfiltration via export	Export actions are permission-gated and logged; large or bulk exports can be flagged for administrator review.
Credential or secret leakage	Centralized secrets manager, no secrets in source control, automatic rotation, and scanning of container images and dependencies in CI.

Threat	Mitigation
Denial of service on search/AI endpoints	Rate limiting at the API gateway, and queued asynchronous processing for bulk document submission.
Tampering with audit or clause data	Tamper-evident audit logging and role restrictions on who can edit Clause Master or override AI output.

11. Design Considerations

This section records the key design decisions behind the architecture in Section 5, the trade-offs involved, and the alternatives that were considered.

Decision	Chosen Approach	Alternatives Considered	Rationale
Vector storage	PostgreSQL with the pgvector extension	Dedicated vector database (e.g., a standalone vector store)	Keeping relational and vector data in one engine avoids operating a second database, simplifies transactional consistency between Doc Content Master rows and their embeddings, and is sufficient at the expected data volumes.
Clause detection without headers	4-page chunking with semantic clause discovery	Fixed-size token chunking; single-pass whole-document extraction	Page-based chunking preserves human-readable page references for traceability (a core usability requirement), which token-based chunking would break; whole-document extraction risks missing clauses in long agreements.
Document processing execution	Asynchronous, queued processing	Synchronous processing in the request path	Contract parsing and LLM extraction can take minutes; synchronous processing would block the UI and does not scale to bulk uploads.
Authentication model	Delegated SSO (SAML/OIDC) only	Platform-managed username/password accounts	Delegated SSO avoids storing credentials, inherits the organization's existing MFA and password policies, and matches NFR-SEC-01.
Service structure	Modular application services (Document, Clause, Search, Dashboard, Admin) behind an API gateway	Single monolithic backend service	Modular services allow the AI processing layer to scale independently from lighter-weight services like Admin, and support independent

Decision	Chosen Approach	Alternatives Considered	Rationale
			deployment (NFR-MAINT-01), at the cost of some added operational complexity versus a monolith.
Clause extraction approach	LLM-based extraction scoped by document type, with Clause Master as the governing configuration	Purely rule-based keyword/regex extraction	An LLM handles varied contract language and non-standard phrasing far better than static rules, though it requires the confidence-score and human-review workflow (Section 3.10) to manage occasional AI error.
Reranking in the query pipeline	Optional reranking step after top-K retrieval	No reranking; rely on vector similarity alone	Reranking improves answer relevance for ambiguous queries at a small latency cost; making it optional lets the system skip it when speed matters more than precision.

DESIGN PRINCIPLE

Each decision above favors traceability and incremental delivery over theoretical completeness: every AI output must be explainable back to a source page, and every component should be replaceable without redesigning the whole platform.

12. Use Case Specifications

The following use cases describe the end-to-end user journey introduced in the BRD in testable form, from platform access through search and export.

UC ID	Use Case	Primary Actor	Description
UC-01	Access Platform	All Users	User logs in via Single Sign-On (SSO) using organizational credentials; role-based access determines available projects, documents and actions.
UC-02	Select Project	Business Teams, Legal Team	User selects the relevant project or repository category where the contract should be stored.
UC-03	Submit Contract	Business Teams	User submits one or more documents; system validates format, size, duplicate status, access rights and project mapping.

UC ID	Use Case	Primary Actor	Description
UC-04	AI Processing	System (automated)	Backend reads the document, extracts structure, identifies document type, detects clauses, creates embeddings and generates summaries.
UC-05	View Contract Snapshot	All Users	User views a business-friendly summary with key terms, important dates, risk indicators and document status.
UC-06	Review Clause Analysis	Legal Team	User reviews extracted clauses with summaries, page references, risk level, source text and confidence score.
UC-07	Validate AI Output	Legal Team	Legal reviewer accepts, corrects, comments on or overrides AI-generated output before final review completion.
UC-08	Monitor Risk	Compliance Team	Users track missing clauses, high-risk language, non-standard terms, upcoming renewals and auto-renewal obligations.
UC-09	Search Repository	All Users	Users search by clause, counterparty, date, document type, value, keyword or natural language query.
UC-10	Export Results	All Users	Users export summaries, search results, risk reports and dashboard data for reporting or review.

13. Acceptance / Validation Criteria

ID	Acceptance Criteria
AC-01	Users can submit supported contract documents under a selected project.
AC-02	System validates document format, duplicate status, size and access permission.
AC-03	ADI/OCR extracts document text, layout, headings, tables and page references.
AC-04	LLM identifies the correct document type.
AC-05	Clauses are detected using section headers when available.
AC-06	Four-page chunking is used when section headers are unavailable.
AC-07	Extracted clauses are stored in Document Content Master.

ID	Acceptance Criteria
AC-08	Document metadata is stored in Document Master.
AC-09	Each extracted clause displays source content and page number.
AC-10	Contract Snapshot displays type, value, auto renewal, governing law, key dates and term.
AC-11	High-risk and missing clauses are clearly highlighted.
AC-12	Users can search contracts by clause, counterparty, date, document type, value or keyword.
AC-13	Users can export summaries, extracted clauses and reports.
AC-14	Legal reviewers can validate, update or override AI output.

14. Risks and Mitigation

Risk	Mitigation
Poor scanned document quality may reduce OCR accuracy.	Flag low-confidence text extraction and route for manual review.
Clause language may vary across legacy agreements.	Use Clause Master, LLM-based semantic matching and embeddings to improve detection.
Missing section headers may affect direct extraction.	Use four-page chunking and semantic clause discovery.
Document type may be misclassified.	Allow Legal Reviewer or Admin to override document type.
Sensitive contract data may be exposed.	Apply role-based access, secure storage, encryption and audit logs.
Clause Master may evolve during implementation.	Start with minimum mandatory clauses and allow version-controlled updates.

15. Glossary

Term	Definition
ADI	Azure Document Intelligence or equivalent document parsing engine used to extract text, layout and structure.
LLM	Large Language Model used for classification, clause identification, summarization and insights.

Term	Definition
Document Master	Database table containing document-level metadata, classification and processing status.
Document Content Master	Database table containing clause-level extracted content, page references, embeddings and summaries.
Clause Master	Admin-managed list of clauses and key terms used as extraction criteria.
Embedding	Vector representation of text used for semantic search and retrieval.
Contract Snapshot	Dashboard view summarizing important contract information.
AI Insights	AI-generated observations related to risk, obligations, missing clauses and review actions.
Chunking	Splitting content into smaller sections, such as four-page chunks, for better clause identification.

FINAL SUMMARY

This SRS defines the functional, data, interface and non-functional requirements needed to build C.L.E.A.R. — converting unstructured contracts into a structured intelligence repository through ADI parsing, LLM classification, clause extraction, semantic search, Contract Snapshot dashboards and AI-generated risk insights.